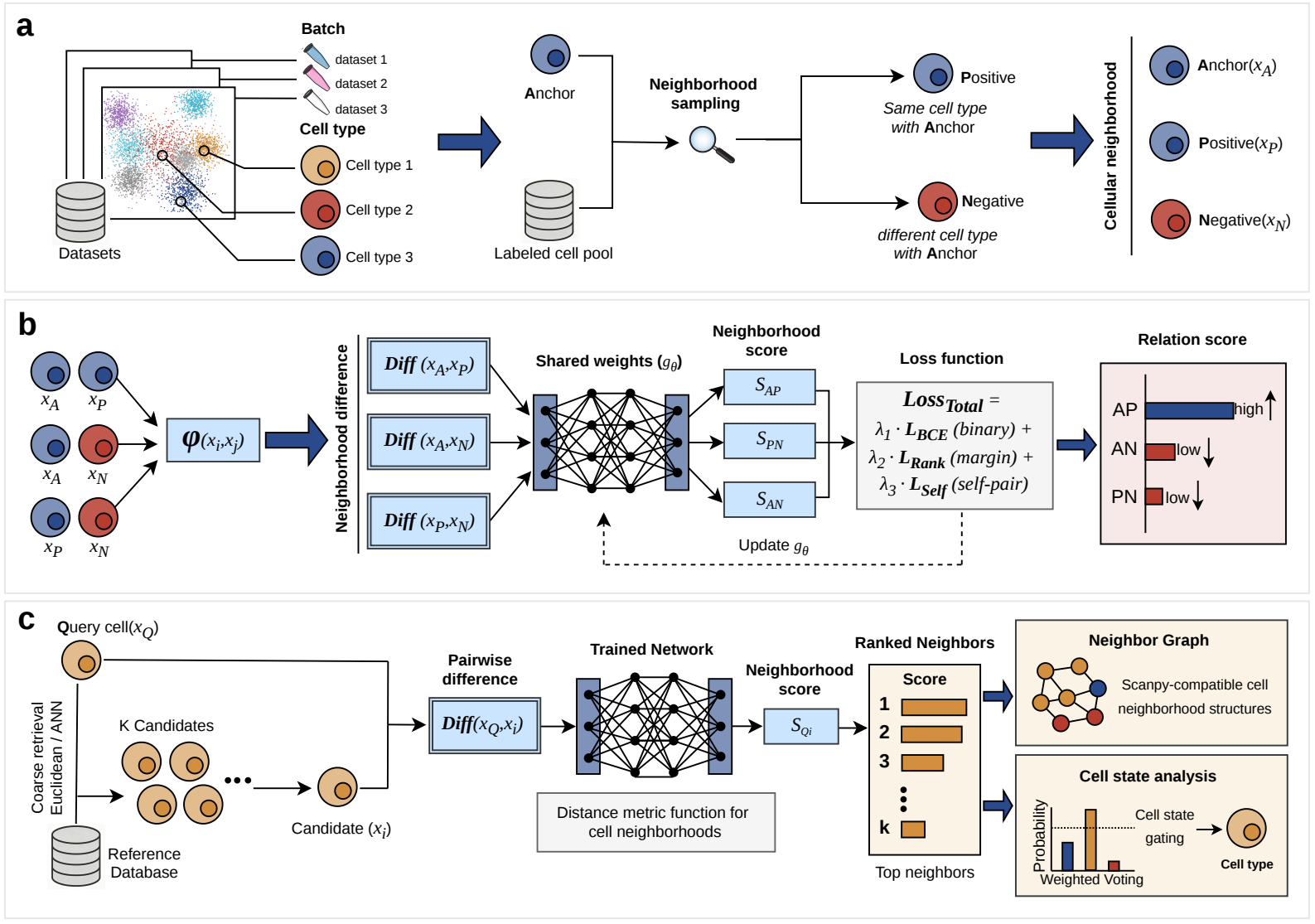

a
Batch
dataset 1
dataset 2
dataset 3
Cell type
Cell type 1
Cell type 2
Cell type 3
Datasets
Anchor
Labeled cell pool
Neighborhood sampling
Positive
Same cell type with Anchor
Negative
different cell type with Anchor
Cellular neighborhood
Anchor(x_A)
Positive(x_P)
Negative(x_N)
b
x_A x_P
x_A x_N
x_P x_N
$\varphi(x_i,x_j)$
Neighborhood difference
Diff (x_A,x_P)
Diff (x_A,x_N)
Diff (x_P,x_N)
Shared weights (g_θ)
Neighborhood score
S_{AP}
S_{PN}
S_{AN}
Loss function
$Loss_{Total}$ =
$\lambda_1 \cdot L_{BCE}$ (binary) +
$\lambda_2 \cdot L_{Rank}$ (margin) +
$\lambda_3 \cdot L_{Self}$ (self-pair)
Update g_θ
Relation score
AP high
AN low
PN low
c
Query cell(x_Q)
Coarse retrieval Euclidean / ANN
K Candidates
Candidate (x_i)
Reference Database
Pairwise difference
Diff(x_Q,x_i)
Trained Network
Distance metric function for cell neighborhoods
Neighborhood score
S_{Qi}
Ranked Neighbors
Score
1
2
3
k
Top neighbors
Neighbor Graph
Scanpy-compatible cell neighborhood structures
Cell state analysis
Probability
Weighted Voting
Cell state gating
Cell type